

Tendon Injuries in the Hand

Robin N. Kamal, Jacob D. Gire

Although not every team physician needs to be a “hand surgeon,” the frequency of hand and wrist injuries caused by sporting activity requires expertise in the diagnosis and evaluation of these injuries. Regardless of the sport, an athlete with a hand injury intuitively appreciates the effect that a hand injury has on function and competitive ability. The injuries often sustained in each individual sport generally have distinguishing characteristics, because most sports place specific demands on an athlete’s hands in a consistently reproducible manner. Accordingly, certain types of athletes are more prone to certain injuries.^{1–4}

Caring for athletes requires an understanding of their preferences for the tradeoffs that exist for various treatments. As treating physicians, surgeons need to appreciate the nuances of patient wishes, risk tolerance, and expectations to guide a shared decision on treatment. Restoring the athlete’s unique skill set after these injuries provides professional challenges that often require consultation with a hand and upper extremity specialist.

Four mechanisms responsible for athletic hand injuries have been proposed by Mirabello and colleagues⁵: throwing, weight bearing, twisting, and impact. Usually the injury is a combination of factors. Werner and Plancher⁶ categorized the potential for injury based on the type of sport. Mechanisms included were impact with a ball or competitor; contact with a racquet, stick, or club; and external contact, as is seen in gymnastics, rock climbing, and weight lifting. Although almost any injury can be seen in any athlete, certain sport-specific patterns of injury have been recognized. For example, interphalangeal (IP) fracture/dislocations are frequent in ball sports such as volleyball and basketball. Hamate fractures are often diagnosed in golfers, baseball hitters, and tennis players. Some injuries are notoriously easy to miss, such as a flexor digitorum profundus (FDP) tendon avulsion in a football or rugby player. Because of the small margin for diagnostic error and the temporal nature of most recommended interventions, neglect and/or misguided treatment algorithms are certain to lead to a poor outcome. Maintaining a high index of suspicion for the worst-case scenario may be the first step in evaluating the injured hand of an athlete.

In general, providing care for an athlete can be challenging for certain populations, such as elite athletes, as expectations and time from play have greater significance. Coaches, managers, agents, and parents may all have various levels of understanding about injuries and disparate expectations for return to play (RTP). At times, these misaligned goals can be counterproductive. Our initial goals are to make an accurate diagnosis, present options

comprehensively, communicate effectively, and create an environment for shared decision-making. The patient management questions set forth by Green and Strickland⁷ create a solid framework for treatment algorithms:

1. Is the method of treatment expected to provide the best long-term result?
2. Would we manage this injury in a similar manner in a nonathlete?
3. Are the potential complications of my treatment significantly greater than might be expected from a more conservative approach?
4. Will the treatment allow the athlete to return to competition with little risk for reinjury?
5. Would reinjury unfavorably influence the prognosis for a satisfactory recovery?

These guidelines apply across the entire scope of sports medicine. Confidentiality, attentive communication, and a patient-centered attitude are all essential to gain the athlete’s trust and maximize outcome.

This chapter focuses on the tendon injuries most commonly encountered by the general orthopaedic surgeon or sports medicine specialist who is acting as first responder. Consequently, only the most pertinent elements of the history and physical examination, followed by key imaging findings and treatment options, are presented here. Clinical pearls, decision-making principles, and our preferred surgical techniques are interspersed within the text. Given that most sports involve intensive use of the hands, both open and closed injuries to the flexor and extensor tendons happen routinely. For open injuries, examination, exploration, and repair of damaged structures are performed expeditiously. Closed tendon injuries in the hand may be subtle and overlooked, but they can cause sufficient morbidity that they deserve special attention. Extensor mechanism injuries are possible at the distal interphalangeal (DIP), proximal interphalangeal (PIP), and metacarpophalangeal (MCP) joint levels, with each producing consistent findings that must be diagnosed by the treating physician. On the flexor side, a FDP avulsion, or “jersey finger,” is a serious and often neglected injury that will compromise composite digital flexion if it is not treated (Fig. 73.1).

MALLET FINGER

The anatomy of the dorsal apparatus of the finger is a complex and intricate structure.^{8–10} Disruption of the attachment of the

Abstract

Hand injuries occur frequently in sport. As such, hand surgery expertise is needed at times for proper evaluation and treatment. Our initial goals are to diagnose accurately, present options comprehensively, communicate effectively, and create an environment for shared decision-making. As treating physicians, surgeons need to appreciate the nuances of patient wishes, risk tolerance, and expectations to guide a shared decision on treatment. This chapter focuses on the osseous and soft tissue injuries most commonly encountered by the general orthopaedic surgeon or sports medicine specialist who is acting as first responder. Consequently, only the most pertinent elements of the history and physical examination, followed by key imaging findings and treatment options, are presented here. Clinical pearls, decision-making principles, our preferred surgical techniques, and post-operative management protocols are interspersed throughout the chapter.

Keywords

hand injuries
tendon injuries
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sport

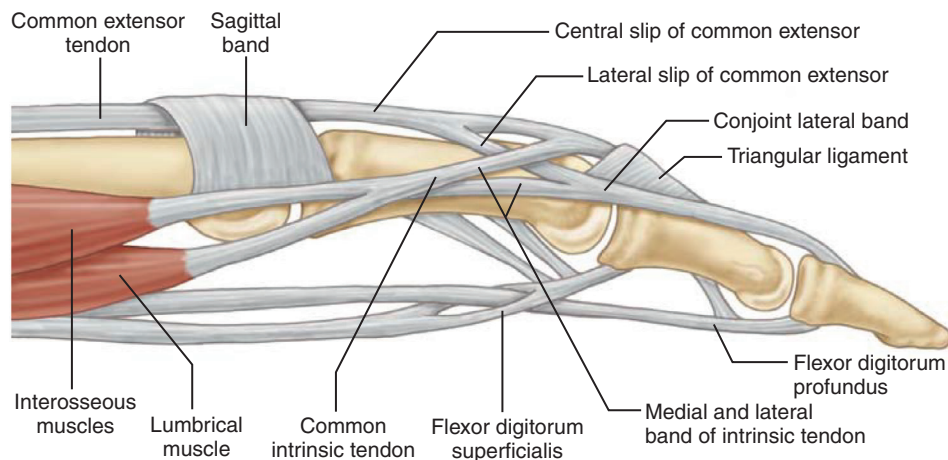


Fig. 73.1 Anatomy of the flexor and extensor apparatus.

terminal extensor tendon into the dorsal base of the distal phalanx has been termed *mallet finger*. Synonyms for this injury are *base-ball finger* and *drop finger*, and “jamming” or “stoving” injuries in ball sports. Axial load and forced flexion of the DIP joint can stretch the terminal tendon, avulse the tendon attachment, or cause an avulsion of bone from the dorsal epiphyseal ridge of the distal phalanx. Warren and associates¹¹ described an area of terminal tendon hypovascularity as the site of vulnerability. The physical examination demonstrates the drooped posture of the DIP joint and the inability to completely extend the joint actively. The degree of dorsal swelling and tenderness is variable. Players may also present with an inability to extend the DIP joint after this injury with PIP joint volar plate laxity with obvious hyperextension and a resulting swan neck deformity. Radiographs are obtained to define any bony injuries, with close attention paid to large avulsion fractures associated with volar subluxation of the DIP joint. It is more common to see a smaller avulsion fracture with variable displacement but without significant joint malalignment.

Extension splinting is the treatment of choice for virtually all mallet finger injuries.^{12–16} A comfortable DIP joint extension splint is placed; hyperextension is avoided because it may lead to dorsal skin necrosis, and the PIP joint is left free to avoid unnecessary proximal joint stiffness. Use of a dorsal, volar, or custom thermoplastic splint produce equivalent results.¹⁷ Splints are worn full time for 6 weeks, with care taken to avoid skin breakdown. Traditionally this has been followed by nighttime splinting; however, a recent randomized controlled trial found that this is unnecessary.¹⁸ Interestingly, as long as the joint is passively supple to full extension, nonoperative treatment may be commenced up to 3 months after the injury was sustained with comparable outcomes.

A rare unstable fracture/dislocation of the DIP joint may warrant surgical treatment (Fig. 73.2). Some have advocated for operative intervention when there is more than one-third of the articular surface involved;^{19,20} however, in the absence of high grade joint subluxation, extension block pinning was not superior to splint immobilization. Instead, the authors reserved operative fixation for those with high grade subluxation, defined as volar displacement of the dorsal cortex of the distal phalanx beyond



Fig. 73.2 Fracture dislocation of the distal interphalangeal joint with joint subluxation.

the central point of the middle phalangeal head on the lateral view.²¹ Complications with surgical treatment have previously been observed in up to 41% of operatively treated injuries.²² The most commonly observed complication was marginal skin necrosis of the distal phalanx, but recurrent flexion deformity, permanent nail deformities, transient infections of the pin tracts, pullout wires or suture, dorsal prominence, and osteomyelitis were also observed.

We collectively treat all but a few select mallet finger injuries nonoperatively. The use of a comfortable volar-based thermoplastic DIP extension splint is preferred. The splint may be further secured with use of 0.5-inch adhesive tape or Coban wrap. A second splint might be provided for wear during showering and should be exchanged on a hard surface that supports the joint in full extension. Splinting is continued for 6 weeks. Athletes who either remove the splint prematurely or inadvertently dislodge and flex the DIP joint must essentially “start the clock over.” As such, agreement on the part of the athlete to partake in the entire length of treatment is necessary prior to initiating treatment. Simultaneous volar and dorsal splinting or digit extension casting can be provided to athletes in contact sports who want to play during the period of immobilization, with the added

risk of maceration, loss of immobilization, and injury to other joints. Inclusion of the entire digit with buddy taping to the adjacent finger should be considered during sport to provide additional stability and protect the adjacent joints.²³

For those that wish to undergo treatment but external splinting is not tolerated, transarticular fixation of the DIP joint in extension with an oblique k-wire buried underneath the skin has been previously utilized²⁴; however, this does place the athlete at risk for pin migration or pin breakage with play.^{23,25} For those that cannot play their sport effectively with the DIP joint in full extension, such as in ball throwing sports, options include immediate splinting with return to sport after 6 weeks, delayed splinting after the season has completed, and no treatment. For many athletes, delayed or no treatment may be preferred and is generally well tolerated and allows immediate return to sport with pain treated symptomatically.²⁵ If electing to delay or forego treatment, the athlete should be made aware of the risk of developing swan neck deformity or functional/cosmetic impairment.

In neglected cases or in those undergoing delayed treatment, splinting is prescribed for a longer period until satisfactory clinical results are achieved. Results are generally acceptable to the patient; however, even with proper treatment, a residual extensor deficit of 20 degrees or more is present in about a third of patients, with greater extensor lag at presentation predictive of worse final function. Fortunately this does not correlate with patient satisfaction with treatment or disability.¹⁸ Persons with PIP volar laxity are fitted with a figure-of-eight-type splint to block PIP hyperextension. Chronic mallet injuries may lead to swan neck deformity, and chronic rigid deformities or those with symptomatic DIP instability are referred to hand specialists for consideration of tenodesis or DIP fusion.

Less common than digit mallet finger injury is the mallet thumb, which is a tear of the extensor pollicis longus (EPL) insertion at the distal phalanx base. The distal phalanx of the thumb is longer, thicker, and wider than the distal phalanges of the digits. In addition, the terminal insertion of the EPL tendon into the dorsal epiphysis of the thumb's distal phalanx is wider and thicker than the terminal tendon insertion in the fingers. The IP joint of the thumb has variability in its arc of motion. Full extension may range from 0 to 60 degrees of hyperextension.

If the presentation is delayed, it is important to obtain the actual injury date from the athlete. As with mallet finger injuries, active extension of the thumb IP joint is markedly less than on the uninjured side. Tenderness at the dorsal prominence of the distal phalanx base is usually present in isolated mallet injuries and helps rule out a proximal EPL tendon lesion. In axial load or contact mechanisms of injury, concomitant fractures, IP joint collateral ligament injuries, and proximal thumb injuries are possible and should be carefully examined. Radiographs should be obtained, and the relationship between the phalanges is noted at the thumb IP joint on the lateral view.

Most published reports of treatment of this injury are limited to small case series. Operative treatment has been recommended by some investigators^{26–28}; however, equivalent results have been documented with closed treatment.^{29–33} Some athletes may elect for operative repair to decrease the amount of full-time immobilization. The fixation of a terminal phalanx avulsion fracture

has similar complications as operative treatment of bony mallet fingers, and caution should be observed.

The athlete may be best served by not playing for at least 6 weeks regardless of the treatment regimen, especially in contact sports. It is virtually impossible to protect the thumb's IP joint while having a functioning thumb ray that can meet the demands of competition. For this reason, some athletes choose to delay definitive management of these injuries until the off-season.

CLOSED BOUTONNIÈRE (CENTRAL SLIP RUPTURE)

The extensor digitorum communis tendon divides into three parts over the PIP joint of the digit. The central slip inserts into the dorsal base of the middle phalanx at the epiphysis. Two lateral slips of the extrinsic extensor tendon separate from the central slip just proximal to the PIP joint and accept a tendinous contribution from the intrinsic musculature to become the conjoined lateral bands at the level of the middle phalanx. These conjoined lateral bands are joined by the oblique retinacular ligament, which facilitates conjugate extension of the PIP and DIP joints. In a closed injury, with forced flexion of the PIP joint, the central slip can rupture from its insertion. Concurrent injury to the tenuous triangular ligament allows volar subluxation of the lateral bands relative to the PIP joint axis of rotation. The boutonnière posture is defined by the resulting flexion moment at the PIP joint and hyperextension moment on the DIP joint from the malpositioned lateral bands (Fig. 73.3). Avulsion fractures at the central slip attachment site can also cause this deformity.³⁴ Boutonnière deformities may be classified as either acute or chronic and as either supple or rigid.

The Elson test has been shown to diagnose central slip ruptures most accurately (Fig. 73.4).^{35,36} The affected digit is flexed 90 degrees over the edge of a table, and the patient is asked to actively extend the PIP joint against resistance. If the central slip is intact, the DIP joint remains supple. If the central slip is ruptured, the DIP joint remains rigid. This test will not diagnose a partial injury to the central tendon and may be inhibited by pain or lack of patient cooperation. As a confirmatory test, the PIP joint is held passively extended and the athlete is asked to flex the DIP joint. The inability to flex the DIP joint indicates a tear of the central slip.

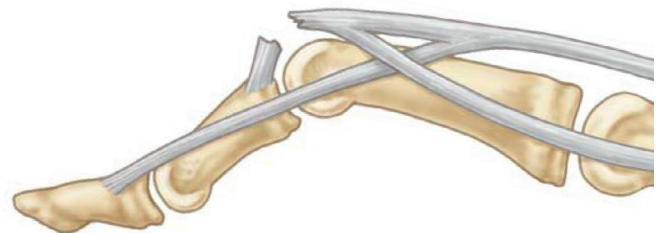


Fig. 73.3 Boutonnière deformity resulting from a central slip rupture with distal interphalangeal joint hyperextension.

Acute open lacerations of the central slip may be repaired with the PIP joint protected with a dynamic extension splint for 6 weeks. Although operative repair of a closed boutonnière deformity has been reported, preferred treatment for most of these deformities is with PIP extension splinting, leaving the DIP joint free (Fig. 73.5).^{37,38} DIP flexion is encouraged because it draws the lateral bands dorsally and thereby promotes an extension force to the PIP joint.^{39–41} Extension splinting with the aid of a hand therapist requires strict attention and must be uninterrupted. If compliance is an issue, percutaneous pinning of the PIP joint is an option, with pin removal at no later than 3 weeks and protected mobilization with an extension splint for an additional 2 to 4 weeks. Return to sports in the nonprofessional athlete at 6 weeks is possible with buddy taping of the finger. In the elite athlete, many factors need to be weighed to arrive at the best treatment for the individual. Some players in certain sports will opt to accept the cosmetic deformity and return to active play without splinting once the pain subsides,

given the stiffness that may develop with nonoperative or operative treatment of the digit.⁴²

For the athlete who presents with severe DIP hyperextension (indicating significant retraction of the central slip), operative repair of the central slip with a suture anchor (Fig. 73.6) and joint pinning for 2 to 3 weeks may be an option. With a neglected or chronic boutonnière deformity, the PIP contracture can be supple or fixed. If passive PIP extension is full, splinting can be tried, unless the athlete is unable to devote the 6 to 8 weeks necessary for successful closed treatment. If the PIP contracture is fixed, active splinting or serial casting is performed to effect full PIP extension. Once full passive extension is attained, static PIP splinting is maintained for 6 weeks. Rarely, release is performed by cutting the check rein ligaments located proximal to the volar plate, which is sometimes combined with open reconstruction of the central slip and mobilization of the lateral bands, or staged reconstruction.⁴³

It should be reinforced that a “pseudoboutonnière” follows a ligamentous injury as a result of a hyperextension mechanism, with resultant scarring and contracture of the volar plate. This mechanism postures the PIP joint in flexion, mimicking a

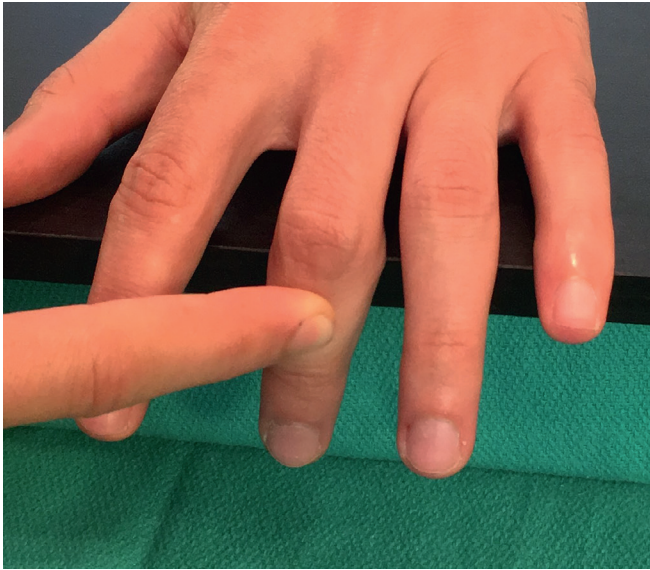


Fig. 73.4 Elson test to evaluate for central slip ruptures.



Fig. 73.5 A proximal interphalangeal extension splint allowing distal interphalangeal flexion for treatment of a boutonnière deformity.

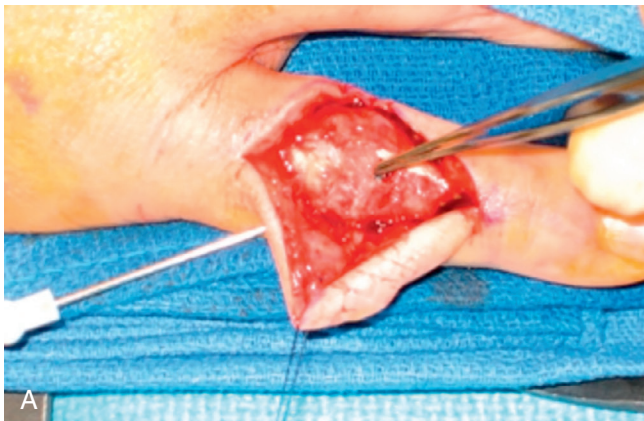


Fig. 73.6 (A) The dorsal approach for repair of a central slip rupture with pinning of the proximal interphalangeal (PIP) joint. (B) Central slip repair with a suture anchor and PIP joint pinning.



Fig. 73.7 Volar plate avulsion from the middle phalanx after hyperextension injury of the proximal interphalangeal joint.

boutonnière position.^{44,45} However, the hyperextension mechanism of injury and subsequent examination can readily distinguish a true boutonnière from a pseudoboutonnière. In the latter, a chip avulsion fracture fragment is often seen near the volar lip of the middle phalanx where the injury occurred (Fig. 73.7).⁴⁶ In addition, with a pseudoboutonnière, the tenderness is almost exclusively at the volar side of the PIP joint, and the DIP motion is essentially normal. Treatment of pseudoboutonnière is directed at progressive stretching of the PIP volar plate, with the introduction of dynamic splinting if necessary.

SAGITTAL BAND RUPTURE

The extensor digitorum communis tendons are maintained over the dorsal apex of the MCP joints by a dorsal sling of adjacent transverse fibers, termed the *sagittal bands* (Fig. 73.8). The sagittal bands act as a tether to prevent either radial or ulnar subluxation of the communis tendon at the level of the MCP joint. The sagittal band invests the extensor mechanism, crossing both volar and dorsal to it, ultimately blending into the volar plate.⁴⁷ Radial sagittal band rupture can occur from forceful finger extension (a “flea-flicker” injury), a direct blow such as in a boxer, or an ulnarly directed force.⁴⁸ The radial sagittal band of the long finger ruptures most commonly, although ulnar sagittal band ruptures have been reported.⁴⁹ Pain and snapping may occur, and a careful examination will document the tendon subluxation or dislocation. Clinically passive extension of the MCP joint is possible and the patient can then usually maintain the finger in an extended position with the extensor tendon centralized; however, the patient may not be able to obtain MCP extension from full flexion. Tendon subluxation is promptly reproduced upon active MCP flexion.

Acute sagittal band ruptures can be treated with a static MCP extension splint, relative motion extension splint, or sagittal band bridge.^{50–52} Preference is a hand based volar MCP extension splint or a relative motion extension splint, continued for approximately 4 to 6 weeks or until central

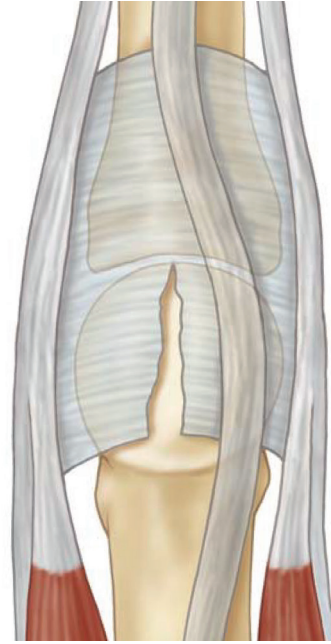


Fig. 73.8 Radial sagittal band tear resulting in ulnar subluxation of the extensor tendon.

extensor tendon tracking over the MCP joint returns when actively making a fist. Treatment of this condition represents a rare instance in which the MCP is immobilized in extension, counter to the “safe” position of the hand where the MCP joints are maximally flexed to prevent collateral ligament scarring in a lengthened position. For chronic cases, a trial of nonoperative treatment is attempted; however, in those with symptomatic persistent dislocation of the extensor tendon, repair may be indicated. One option includes primary repair of the torn sagittal band with realignment of the extensor tendon by releasing the opposing sagittal band.^{53–56} Alternatively, various reconstructive techniques have been described that utilize adjacent structures to restrain the extensor tendon. These techniques include use of adjacent juncturae, anomalous tendons (e.g., extensor medius proprius or extensor indicis et medii communis), palmaris autograft, or distally based slips of the injured extensor tendon to create a restraint to tendon subluxation.^{57–62}

The sagittal bands are approached via a curvilinear dorsal incision and torn fibers are primarily repaired. If there is redundant tissue, such as in chronic scenarios, this is imbricated and the opposing sagittal band and/or junctura may necessitate release to allow for centralization of the tendon. The ulnar junctura tendinum can be flipped over and sutured to the palmar radial sagittal band remnant of the deep intermetacarpal ligament to provide additional restraint against ulnar subluxation.⁵⁹ Alternatively, the ulnar slip of EDC can be cut proximally, maintaining its distal insertion and routed around the extensor tendon and radial collateral ligament and sutured to itself. Post repair, a relative motion splint may be utilized to facilitate early ROM.^{63,64} RTP is delayed for 8 to 12 weeks post-op to allow adequate soft tissue healing.

Few athletes will be comfortable enough to continue competitive play in the acute phase. This situation, coupled with the possibility of further damage to the extensor hood or underlying dorsal capsule from repetitive trauma, leads to the decision to provide treatment in-season in the majority of cases. Mild injuries without subluxation may require only a short period of missed play, with buddy taping utilized upon return, whereas more severe injuries may prevent athletes from competing for several months. We are not aware of ways to manage this pathology safely and effectively while allowing athletes (such as baseball or basketball players) to compete at a high level for the remainder of a season as they await out-of-season surgery. The orthosis limits functional RTP and places the hand at risk for further injury.

Although the hands of any boxer can be subjected to a multitude of injuries, *boxer's knuckle* specifically refers to chronic attritional disruption of the dorsal anatomy at the MCP joint. This disruption may include the central extensor tendon, sagittal band, and/or dorsal capsule.^{53,65} A large capsular rent may allow for metacarpal head protrusion and persistent flow of synovial fluid from the joint cavity to the subcutaneous space. Metacarpal head osteochondral defects may be present. Athletes who engage in punching and have early symptoms may possibly continue if proper protection is used and supervision is provided. Operative treatment is comparable to sagittal band repair with repair, reconstruction, or débridement of the dorsal capsule also performed.^{53,58,65} As dorsal capsular repairs may lead to a loss of MCP flexion, excisional débridement of the dorsal capsule when a rent is encountered leaving the capsule open is preferred.⁶⁶ Post operatively, immobilization is brief (2 to 3 weeks), with the MCP joint flexed 60 degrees to minimize loss of joint flexion. Wrapping and gloving strategies with shock-absorbing padding may decrease further injury, but competition rules may restrict the use of some of the more effective types of equipment for these athletes.

JERSEY FINGER

"Jersey finger" is an avulsion of the FDP from its insertion on the distal phalanx (Fig. 73.9). This injury classically occurs when an athlete, such as a football or rugby player, forcefully grabs an opponent's jersey with flexed fingers as the opponent duly escapes the grasp. Various theories have been proposed to explain why the ring finger, as opposed to the long finger, is most commonly affected.⁶⁷ The strength of the FDP insertion of the ring finger is significantly less than that of the adjacent digits.⁶⁸ As one grips, the distal segment of the ring finger projects farther and becomes more prominent because of the increased mobility at the ring finger carpometacarpal (CMC) joint.⁶⁹

Leddy and Packer⁷⁰ described three types of injury, based on the following factors: (1) the presence or absence of a bony fragment radiographically, (2) the level to which the tendon retracted, and (3) the status of the blood supply of the avulsed tendon. The tendon can avulse with or without a bony fragment. The level of the bony fragment as shown on a lateral radiograph does not reliably predict the level of tendon retraction because the tendon end and bone fragment may separate.⁷¹ In a type I injury, the tendon has retracted into the palm, without the



Fig. 73.9 Jersey finger injury results in loss of distal interphalangeal flexion.

presence of a bony fragment. Repair within a week is required for this injury because retraction of the tendon into the palm and loss of blood supply lead to contraction. Type II avulsions often are accompanied by a small bone fragment and retract only to the level of the PIP joint, and the vinculum longum may still be intact. This injury can be successfully repaired up to 6 weeks after the injury occurs because the tendon length has been maintained. Type III injuries are accompanied by a large bony fragment and are restrained distally by the A-4 pulley (Fig. 73.10). These injuries also can be successfully repaired weeks after the injury is sustained. After the original three-part description by Leddy and Packer in 1977, a fourth type was added to the classification scheme. This fourth type involves an intra-articular fracture of the distal phalanx and separation of the tendon from the fracture fragment (see Fig. 73.10).⁷²

Loss of isolated DIP joint flexion is the *sine qua non* of this injury and is tested by blocking the PIP joint in an extended position while profundus function is checked (Fig. 73.11A). The middle, ring, and small finger profundus tendons are connected to a shared muscle belly in the forearm, whereas the index profundus typically has its own separate muscle belly. Examination may reveal a prominence where the proximal ruptured tendon end is located. For instance, fullness and tenderness at the A-1 pulley area may indicate a type I injury with retraction to the palm level. Radiographs are important and help determine whether there is an associated fracture or avulsion. Ultrasonography and magnetic resonance imaging (MRI) may be useful adjuncts in equivocal cases, in which the timing of surgery may be dictated by the classification of injury (Fig. 73.12).

Surgical repair is best carried out within a week of the injury, especially for type I injuries where the FDP is retracted into the palm and any further delay may limit the ability to restore the FDP back to its insertion on the distal phalanx. Because it is not

possible to know the exact position of the avulsed tendon end by examination alone, proceeding with acute repair of all jersey fingers in athletes is justified.⁷³ The distal FDP insertion site is exposed with either a Bruner incision or use of the midaxial approach. If the tendon has migrated proximally into the palm, the pulley system and fibro-osseous canal must be maximally

preserved. The tendon can be passed distally with a pediatric feeding tube as a leader or by “milking” the tendon distally with smooth forceps from the palm just proximal to the level of the A-1 pulley. Distal insertional repair was historically performed with a pullout suture, although suture anchor fixation has gained popularity.⁷⁴ More recently, an all inside technique has been

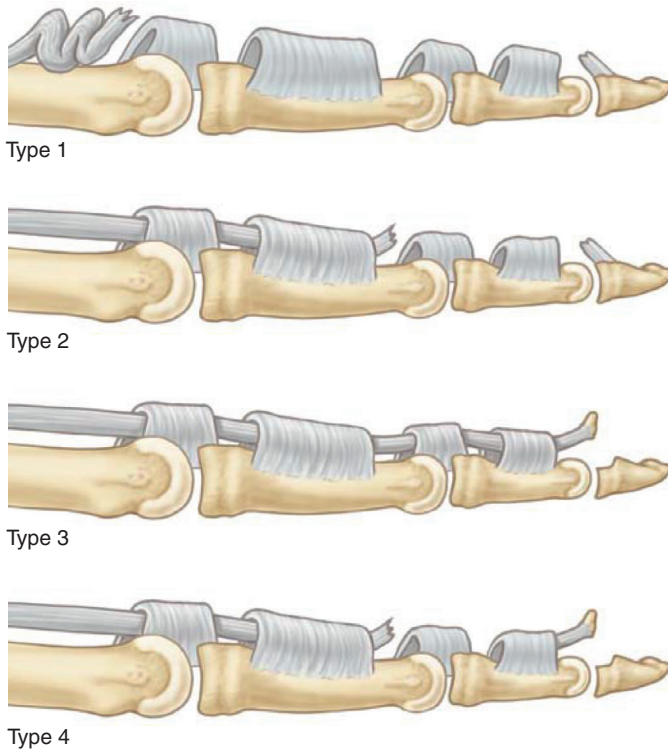


Fig. 73.10 Classification of zone 1 flexor digitorum profundus injury.

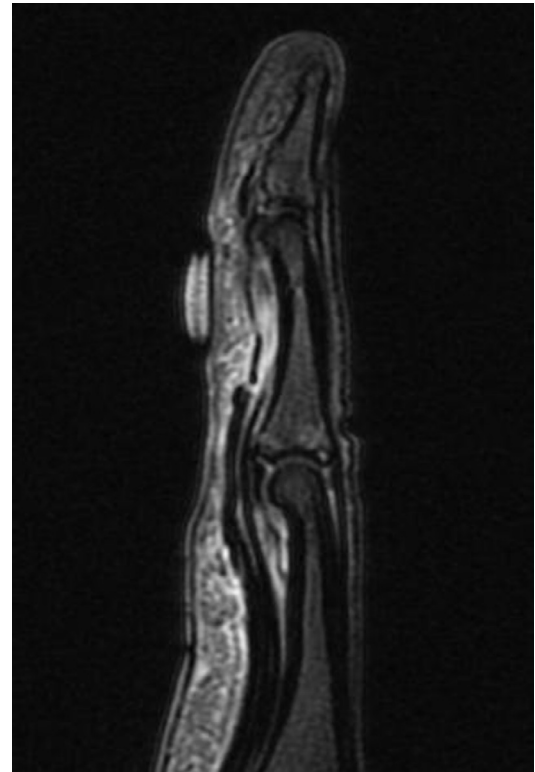


Fig. 73.12 T2-weighted sagittal image of the long finger after injury demonstrating flexor digitorum profundus rupture with retraction to the A2 pulley.

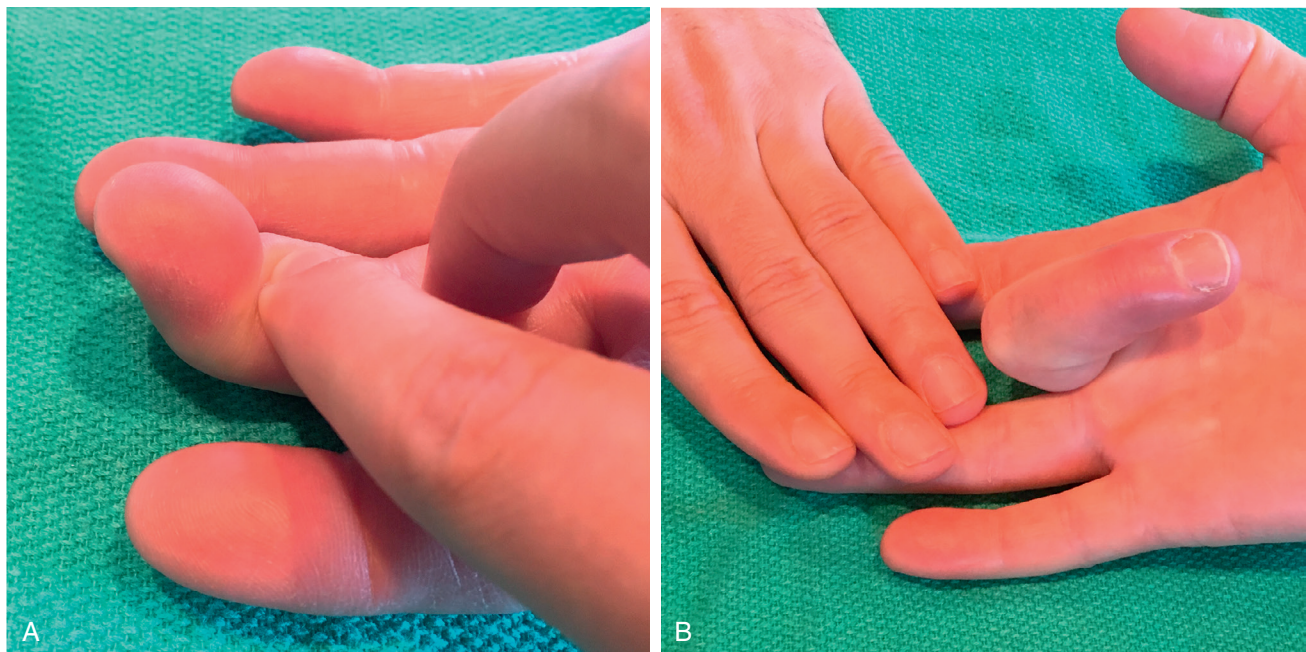


Fig. 73.11 Examination of the (A) flexor digitorum profundus and (B) flexor digitorum superficialis tendons.

described, with the FDP tendon attached to bone with two 3-0 nonabsorbable sutures passed through the phalanx and tied over the dorsal aspect of distal phalanx.⁷⁵ There is wide variation in described techniques, none of which have emerged superior.⁷⁶ If additional fracture fragments are present, they are fixed with mini screws (Fig. 73.13B) before tendon repair. Utilization of the wide awake local anesthesia—no tourniquet technique allows for intraoperative evaluation of repair and may lead toward increased use of earlier active ROM protocols.⁷⁷ After repair, protective dorsal block splinting is provided and a passive Duran flexor tendon protocol is initiated under the supervision of an occupational therapist for the first 4 to 6 weeks. Graduated active flexion is encouraged beginning with place and holds at 2 to 3 weeks. Resistive grasp and return to sports are delayed until 10 to 12 weeks after the repair is made.

Players may not seek immediate treatment for an FDP avulsion injury for a number of reasons. They may believe it is a minor injury that can be treated after the season is over, or they may not want to miss playing for a large part of their season. Informing the player of the long-term consequences of neglecting this injury is imperative, and the need for detailed documentation of conversations with the player cannot be overemphasized. Treating a neglected FDP avulsion is never as satisfying as performing a

primary repair, and the clinical results are not nearly as functional. The following treatment options can be considered for a neglected FDP avulsion:

1. Late reinsertion: Late reinsertion is unlikely to be possible if the tendon has retracted a significant amount and the time from injury is beyond 6 weeks.
2. Acute single-stage reconstruction with a tendon graft: This option is fraught with potential complications, including loss of additional ROM (in the PIP joint), scarring, and worsening of FDP function.
3. Two-stage flexor tendon reconstruction: This approach requires a large amount of effort by the patient, surgeon, and therapist and will probably be considered too aggressive a course to regain a small amount of DIP joint flexion. The first stage includes placement of a silicone rod to reconstitute the flexor tendon sheath, followed by exchange for a tendon graft at least 3 months later. A third flexor tenolysis surgery is probable in this treatment course as well.
4. Stabilization of the DIP joint by fusion or capsulodesis: Stabilization is often the most acceptable choice for the patient when risks and benefits are carefully considered.
5. Excision: If a retracted tendon is a tender mass in the palm, simple excision of the scarred proximal tendon end may relieve discomfort with grasping activities.
6. Nonsurgical treatment: Some players choose to do nothing after weighing the pros and cons of the aforementioned possibilities, and the desire to avoid a possible threat to PIP function is not entirely inappropriate.

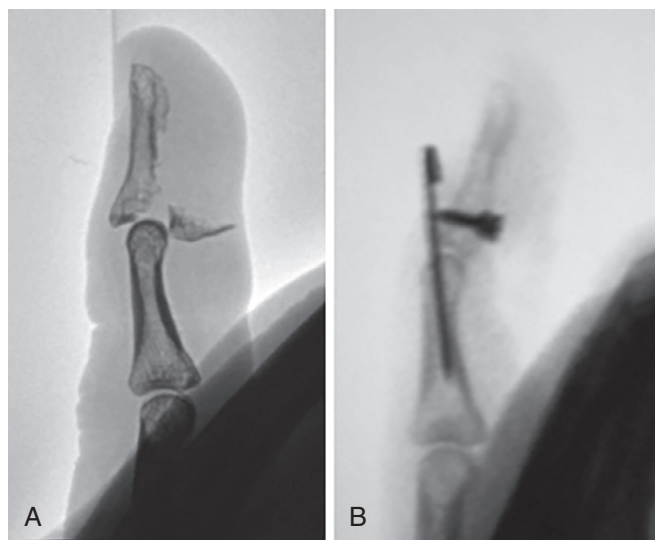


Fig. 73.13 (A) A flexor digitorum profundus avulsion with fracture of the distal phalanx. (B) Fracture fixation and pinning of distal interphalangeal joint subluxation.

PULLEY INJURY

The fibro-osseous canal and system of pulleys within the digits were elegantly described by Doyle and Blythe⁷⁸ in 1977 (Fig. 73.14). Biomechanically the A-2 and A-4 pulleys are critical for normal flexor tendon function and the prevention of “bowstringing.”^{79,80} The A-2 pulley overlies the proximal aspect of the proximal phalanx, whereas the A-4 pulley is at the level of the middle phalanx. Attenuation or frank rupture of a digital pulley occurs most often in rock climbers and baseball pitchers^{81–83} as a result of acute or chronic exposure to forceful contraction of the FDP tendon against a greater than physiologic load. In “free” rock climbers, the flexed DIP joint incredibly supports body weight repetitively and for prolonged intervals of the activity. The rock climbing hand posture of “crimping” specifically places significant strain on the distal part of the A-2 pulley and may understandably

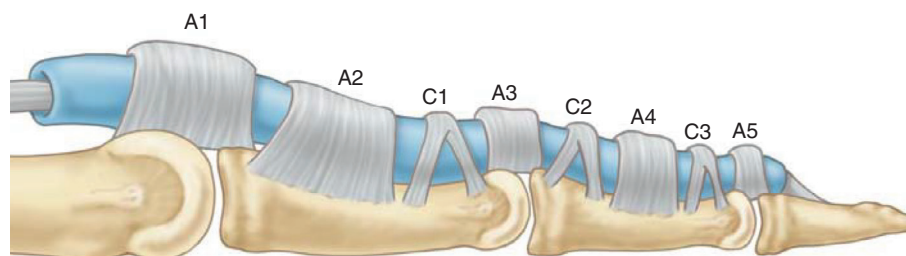


Fig. 73.14 Anatomy of the normal flexor pulley system with associated annular (A) and cruciate (C) pulleys labeled.

result in pulley rupture.⁸⁴ Injury to the thumb pulley system is exceedingly rare. The thumb anatomy differs from the digits, with four components consisting of three annular pulleys, the proximal A-1 at the MCP joint, and distal A-2, which attaches to the IP volar plate and the variable annular pulley. Between the variable and distal annular pulleys, the oblique pulley overlies the proximal phalanx.⁸⁵ Bowstringing can occur if both the A-1 and oblique pulleys are ruptured.⁸⁶

An athlete may complain of an audible pop or tearing sensation when pulley rupture occurs. On clinical examination, the patient reports pain over the flexor sheath. Swelling over the pulley may be present, and loss of terminal DIP flexion may be noted. Discomfort often occurs with active digit flexion. The athlete may perceive weakness, and pitchers will lose fastball velocity. The spectrum of injury ranges from partial injury of a pulley to complete rupture of multiple pulleys with overt bowstringing of the flexor tendons. Rarely both the A-2 and A-4 pulleys fail simultaneously. MRI can aid in confirming the diagnosis and may show edema within the flexor tendon sheath and tendon bowstringing when the images are obtained with applied digital resistance (Fig. 73.15).⁸⁷ Ultrasound may also be utilized to evaluate for pulley rupture, which also allows for dynamic evaluation of the flexor tendon system.⁸⁸

Grading systems for pulley injury have been developed and may help guide treatment.⁸⁹ Grade 1 injuries are pulley sprains, and do not require immobilization. Return to sport may be allowed after 4 weeks, with symptoms initially managed with ice and antiinflammatory medications. Complete rupture of the A4 or partial injury of A2 or A3 represent grade 2 injuries and are treated with brief immobilization, symptom management

as with grade 1 injuries, pulley support taping, and gentle ROM as edema decreases. Corticosteroid injection is avoided, as it may precipitate pulley rupture and delay healing. Initiation of easy sport specific activities is allowed after 4 weeks.⁹⁰ For grade 3 injury with complete rupture of the A-2 and A-3 pulley, an occupational therapist may fashion an external pulley ring for use in addition to support tape to relieve stress on the pulley system. Return to sport is allowed after 6 weeks, with utilization of digit taping in addition to the pulley ring to prevent skin maceration and reduce ring translation.^{23,42} Return of complete ROM is frequently delayed. PIP flexion contractures can complicate the postinjury course.

With grade 4 injuries where rupture of the entire pulley system has occurred, loss of motion and bowstringing are evident. This may require surgical reconstruction of the A-2 and/or A-4 pulleys if symptomatic. The principles of reconstruction are to maintain the flexor tendons near the PIP and DIP centers of rotation. Reconstructed pulleys must be sufficiently strong to allow for early mobilization. A tendon autograft (e.g., the palmaris longus, plantaris, or partial flexor digitorum superficialis) or dorsal wrist retinaculum is used to reconstruct the pulleys.^{91,92} These may be encircling or nonencircling reconstructions, with looped encircling reconstructions shown to be mechanically superior.⁹³ We utilize a looped reconstruction with palmaris autograft for most cases. All scar posterior to the tendon is excised. At the A-2 pulley level, the graft is placed beneath the extensor mechanism and then three separate loops envelop the flexor tendons.⁹⁴ At the A-4 level, however, the graft is placed over the extensor apparatus with two separate loops. Careful intraoperative tensioning is important, as over tensioning can result in loss of flexion and stiffness, whereas under tensioning can result in failure to improve function. Utilization of wide awake local anesthesia techniques, without tourniquet,⁹⁵ can facilitate appropriate tensioning and tendon tracking can be observed intraoperatively with active movement. Postoperatively, passive tendon gliding protocols with place and holds are used, and ring splints help minimize stress to the newly reconstructed pulleys. Full return to sport after reconstruction is delayed for 3 to 6 months, and pulley ring or pulley taping may be necessary for up to 6 months.⁹² Reoperation after pulley reconstruction has been reported at 6%, most commonly for tenolysis.⁹⁶ Other reported complications include stiffness, flexion contracture, synovitis, ischemia of the phalanx resulting in fracture after encircling techniques, and pulley rerupture.^{97,98}

TENDON INJURY COMPLICATIONS

Generally speaking, complications after hand surgery are minimized by careful attention to treatment principles, meticulous surgical technique, and appropriate postoperative rehabilitation. Because small joint stiffness is a pervasive problem throughout the spectrum of hand injury,⁹⁹ prolonged immobilization beyond a couple of weeks is usually not recommended. Techniques or strategies to initiate early motion recovery are advocated after most injuries except for specific closed disruptions of the extensor mechanism. Tendon repair can be complicated by adhesion formation, tendon rerupture, joint contracture, and quadriga.



Fig. 73.15 (A) Sagittal proton density fat suppressed and (B) axial T2-weighted magnetic resonance images demonstrating rupture of the A2 pulley with increased bone to tendon distance.

Some of these complications may decrease as wide-awake flexor tendon surgery is popularized and direct dynamic assessment of tension after repair is performed.⁷⁷

Missed injuries have variable consequences. For example, a chronic nonunion of the hamate hook may require nothing more than simple excision once it is detected, with relief of pain and RTP that is typically fast and predictable. On the other hand, mechanical abrasion of the adjacent ulnar-sided digital flexor tendons by the fragment could lead to rupture and the need for surgical reconstruction, which has huge implications with regard to early restrictions, rehabilitation time, and overall recovery of grasping strength. Missing a jersey finger injury has fewer consequences when the proximal tendon end is discovered close to its insertion rather than retracted into the palm.

Postoperative infection after ambulatory hand surgery is rare. In a retrospective review of almost 9000 outpatient hand surgery cases, the overall surgical site infection rate was 0.35%.¹⁰⁰ Prophylactic preincision antibiotics are typically administered only in prolonged cases or when an implant is used. Risk factors for poor wound healing, such as the presence of diabetes mellitus or heavy tobacco use, may also warrant administration of antibiotics.

Masterful knowledge of both normal and aberrant anatomy, as well as careful surgical dissection, is necessary to avoid iatrogenic injury to cutaneous nerve branches found in the hand, which may lead to dysesthesia, formation of a neuroma, failure to progress with rehabilitation, and a potential second surgery in recalcitrant cases. For example, meticulous dorsal dissection during extensor tendon repair is required to prevent injury to the traversing cutaneous nerves. In our experience, this type of close attention to detail may provide the difference between a stiff, painful hand leading to secondary neuroma excision and tenolysis, and a hand that regains nearly immediate full ROM without pain and a prompt return to sport after fracture healing.

FUTURE CONSIDERATIONS

The future of hand surgery as it pertains to the athlete will likely continue to optimize the art of diagnosis, treatment, and rehabilitation to expedite the quickest and safest return to sport. The in-office use of ultrasound techniques by the practicing surgeon has the potential to facilitate the diagnosis of closed tendon injuries, thus avoiding use of expensive MRI scans and possibly altering treatment plans in some circumstances. Emerging technologies

will continue to advance surgical fixation methods, some of which may offer a level of strength that allows for early mobilization and swifter RTP. Increased use of wide awake surgery will allow for intraoperative dynamic evaluation of tendon repairs, with immediate correction of any observed gaping. Better treatment methods will only be as good as the rehabilitative protocols that follow, challenging our surgeons and therapists to create strategies that safely accelerate the RTP process.

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For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

The following collection of articles represents some of the most comprehensive and well-written expert reviews regarding tendon injuries in athletes.

Citation:

Netscher DT, Pham DT, Staines KG. Finger injuries in ball sports. *Hand Clin.* 2017;33:119–139.

Level of Evidence:

V

Citation:

Freilich AM. Evaluation and treatment of jersey finger and pulley injuries in athletes. *Clin Sports Med.* 2015;34:151–166.

Level of Evidence:

V

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Marino JT, Lourie GM. Boutonnière and pulley rupture in elite athletes. *Hand Clin.* 2012;28:437–445.

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McMurtry JT, Isaacs J. Extensor tendons injuries. *Clin Sports Med.* 2015;34:167–180.

Level of Evidence:

V

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